

Abstract Versions for SPECTRUM

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Five candidate abstracts for 01_slr_forecast_intervention2026.tex. All open with the same three sentences as the manuscript abstract and carry the same core results. They differ in how they handle the slow-versus-fast West Antarctic contrast, which is stated in plain language throughout; none uses the S1/S2 scenario notation or the mixture weight $p(S1)$, since neither is defined until the body.

The handling in each: (1) states the two futures and gives no blended number; (2) leads with the blended number and discloses the weighting behind it; (3) makes the gap between the two futures the headline result; (4) casts the contrast as two independent dimensions of risk; (5) frames the contrast as a question that observation can answer.

Version 1: Two futures, no blended number

Every centimeter of sea-level rise exposes millions more people to annual coastal flooding, yet projections of sea level come with large uncertainties. Many of the commonly used projections rely on complex models calibrated on data too sparse to adequately constrain them. Here, we construct a data-driven sea-level projection framework by calibrating parsimonious physical models for individual sea-level contributors against decades-long independent observational records. We call this framework SPECTRUM. Global mean sea level (GMSL) is withheld from every component calibration and reserved as a test of the assembled framework, which reproduces the satellite-era rate of rise and matches reconstructions within error back to at least 1940, spanning approximately 1 °C of warming. Driven forward, the forecast divides into two futures that the observational record cannot yet distinguish. Should West Antarctic ice loss continue near its observed pace, GMSL rise is predictable and responds to emissions. Holding warming to 2 °C rather than 4 °C avoids 36 cm of rise, and the chance of exceeding 1 meter above pre-industrial falls to near zero below 3 °C. Should West Antarctic retreat instead become self-sustaining, the upper tail opens and responds weakly to emissions. At the 5% exceedance level under 3 °C of warming, the two futures differ by 0.95 m of rise, 177 million people exposed to annual coastal flooding, and US\$127 billion per year in adaptation costs. Which future we are in is the dominant open question in 21st-century sea-level projection.

Version 2: Blended number with the weighting disclosed

Every centimeter of sea-level rise exposes millions more people to annual coastal flooding, yet projections of sea level come with large uncertainties. Many of the commonly used projections rely on complex models calibrated on data too sparse to adequately constrain them. Here, we construct a data-driven sea-level projection framework by calibrating parsimonious physical models for individual sea-level contributors against decades-long independent observational records. We call this framework SPECTRUM. Global mean sea level (GMSL) is withheld from calibration, and the assembled framework reproduces the satellite-era rate of rise and matches reconstructions within error back to at least 1940, spanning approximately 1 °C of warming. The observational record cannot yet distinguish continued West Antarctic loss at its observed pace from a self-sustaining retreat, so we carry both futures and weight the forecast toward accelerating loss. Under that weighting, and averaged over the policy-relevant warming trajectories, GMSL has a 53% probability of exceeding 1 meter above pre-industrial by 2100 and 4.5% of exceeding 2 meters. These odds are a direct consequence of how the two futures are weighted, and they move accordingly. Were West Antarctic loss instead to stay near its observed pace, the chance of exceeding 1 meter above pre-industrial would fall to near zero below 3 °C of warming. West Antarctic dynamics account for approximately 99% of the forecast variance, so emissions reduction lowers the median while leaving the high-risk tail largely unchanged.

Version 3: The gap between the two futures is the result

Every centimeter of sea-level rise exposes millions more people to annual coastal flooding, yet projections of sea level come with large uncertainties. Many of the commonly used projections rely on complex models calibrated on data too sparse to adequately constrain them. Here, we construct a data-driven sea-level projection framework by calibrating parsimonious physical models for individual sea-level contributors against decades-long independent observational records. We call this framework SPECTRUM. Global mean sea level (GMSL) is withheld from every component calibration, and the assembled framework reproduces the satellite-era rate of rise and matches reconstructions within error back to at least 1940, spanning approximately 1 °C of warming. Applied forward, the framework returns a result better expressed as a gap than as a number. Whether West Antarctica continues to lose ice near its observed pace or enters a self-sustaining retreat changes 2100 sea level, at the 5% exceedance level that coastal design uses, by 0.95 m under 3 °C of warming. That single unresolved question separates 177 million people from annual coastal flood exposure, US\$127 billion per year in adaptation costs, and US\$12.7 trillion per year in flood damages absent further adaptation. It accounts for approximately 99% of the forecast variance, while every other contributor to sea level, calibrated here against decades of observations, together accounts for the remainder. Emissions reduction compresses that remainder and lowers the median, but moves the probability of exceeding 2 meters above pre-industrial only from 6% to 3%. The gap is not closed by mitigation. It is closed by finding out which future we are in.

Version 4: Two independent dimensions of risk

Every centimeter of sea-level rise exposes millions more people to annual coastal flooding, yet projections of sea level come with large uncertainties. Many of the commonly used projections rely on complex models calibrated on data too sparse to adequately constrain them. Here, we construct a data-driven sea-level projection framework by calibrating parsimonious physical models for individual sea-level contributors against decades-long independent observational records. We call this framework SPECTRUM. Global mean sea level (GMSL) is withheld from calibration, and the assembled framework reproduces the satellite-era rate of rise and matches reconstructions within error back to at least 1940, spanning approximately 1 °C of warming. Decomposing the forecast separates sea-level risk into two dimensions that respond to different actions. The first is set by ocean thermal expansion, mountain glaciers, and the Greenland Ice Sheet, which track temperature and are therefore compressible by emissions reduction. Were West Antarctic loss to continue near its observed pace, this dimension would govern, and holding warming to 2 °C rather than 4 °C would avoid 36 cm of rise, 61 million people exposed, and US\$49 billion per year in adaptation costs. The second dimension is the possibility that West Antarctic retreat becomes self-sustaining. It accounts for approximately 99% of the forecast variance, sets the upper tail, and responds weakly to emissions: cutting warming from 4 to 2 °C moves the probability of exceeding 2 meters above pre-industrial only from 6% to 3%. Emissions policy governs how much rise is likely. West Antarctica governs how bad the worst case becomes.

Version 5: A question observation can answer

Every centimeter of sea-level rise exposes millions more people to annual coastal flooding, yet projections of sea level come with large uncertainties. Many of the commonly used projections rely on complex models calibrated on data too sparse to adequately constrain them. Here, we construct a data-driven sea-level projection framework by calibrating parsimonious physical models for individual sea-level contributors against decades-long independent observational records. We call this framework SPECTRUM. Global mean sea level (GMSL) is withheld from calibration, and the assembled framework reproduces the satellite-era rate of rise and matches reconstructions within error back to at least 1940, spanning approximately 1 °C of warming. Because every contributor is calibrated separately, the forecast localizes its own uncertainty. Approximately 99% of the variance traces to a single unresolved question: whether West Antarctica continues to lose ice near its observed pace or enters a self-sustaining retreat. The two futures differ by 0.95 m of 2100 rise at the 5% exceedance level under 3 °C of warming, and emissions reduction does not separate them, moving the probability of exceeding 2 meters above pre-industrial only from 6% to 3%. Resolving the question is worth on the order of US\$5 trillion in avoided adaptation costs, with that value largest when the answer arrives early enough to shape infrastructure decisions. The answer is observable. The rate of West Antarctic ice loss over the next two decades would narrow this forecast more than any other measurement, and sustained observation of West Antarctica is what would tell coastal communities how much rise to prepare for.